

INVENTIVE CONCEPTS SUMMARY AND CLAIM SUPPORT FRAMEWORK

Purpose of Inventive Concepts Summary

This section summarizes representative inventive concepts supported throughout the disclosure. The following concepts are provided to demonstrate breadth of inventive contribution and to support future claim drafting. The concepts described are non-limiting and may be combined, subdivided, or expanded during preparation of one or more non-provisional patent applications.

Inventive Concept 1 — Longitudinal Wearable Rigidity Monitoring

A wearable device configured to monitor penile physiology longitudinally using mechanical, thermal, optical, magnetic, and inertial sensing modalities operating independently or through multi-modal sensor fusion.

Inventive Concept 2 — Inferred Physiologic Rigidity Modeling

Computation of rigidity or physiologic state through inferred modeling rather than direct measurement, including compliance modeling, deformation analysis, and adaptive interpretation frameworks.

Inventive Concept 3 — Multi-Modal Sensor Fusion Architecture

Integration of heterogeneous sensing channels into unified physiologic interpretation systems capable of generating composite indices, confidence scores, and derived metrics.

Inventive Concept 4 — Adaptive Longitudinal Personalization

Creation of individualized physiologic baselines and adaptive learning models capable of refining interpretation parameters across extended monitoring periods.

Inventive Concept 5 — Context-Aware Physiologic Interpretation

Use of inertial sensing, environmental context, and behavioral information to improve accuracy of physiologic state classification and reduce artifact influence.

Inventive Concept 6 — Privacy-Preserving Abstract Output Representation

Transformation of physiologic measurements into non-explicit indices, symbolic profiles, or abstract visual representations suitable for consumer, clinical, or social platforms.

Inventive Concept 7 — Multi-Node Distributed Physiology Platform

Operation of the wearable device as a node within a distributed physiologic monitoring ecosystem supporting cross-device synchronization and comparative analysis.

Inventive Concept 8 — Multi-User Comparative Physiology

Correlation and synchronization of physiologic datasets across multiple users to enable interaction modeling, compatibility assessment, or comparative physiologic analytics.

Inventive Concept 9 — Future-Proof Wearable Architecture

Implementation flexibility allowing evolution of form factor, sensing technology, energy systems, communication methods, and computational intelligence while preserving core inventive principles.

The inventive concepts summarized herein provide support for a plurality of future claims, continuations, divisionals, and related patent filings derived from the disclosed longitudinal physiologic monitoring system.